

KARAN MANICKAM

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Education

KSR Institute for Engineering and Technology, Namakkal

Nov. 2022 – May 2026

B.E. Computer Science and Engineering | CGPA: 8.77

Namakkal, Tamil Nadu

- Awarded **Best Outgoing Student** – Department of Computer Science and Engineering (2026 Batch).

Relevant Coursework

- Data Structures
- Database Management
- Computer Networks
- Web Technologies
- Operating Systems
- Software Engineering
- Artificial Intelligence

Experience

Mphasis Limited

Jan 2026 – Apr 2026

Associate Software Engineer Intern

Bangalore, Karnataka

- Developed a Jira-TestRail Workstation using Python, FastAPI, and Flask to identify unlinked test cases and streamline Jira ticket creation, processing and managing 10,000+ TestRail and Jira records.
- Built an AI-powered Jira Assistant using Python, RAG, LangChain, Ollama, and Mistral, enabling natural language queries on Jira data and achieving response times under 3 seconds.
- Contributed to 6+ client Proof of Concepts (PoCs), developing and deploying solutions on local servers while collaborating with core engineering teams and stakeholders for successful delivery.

Praskla Technology

Oct 2025 – Jan 2026

Software Engineer Intern

Namakkal, Tamilnadu

- Developed scalable backend services for an e-commerce platform using Node.js, Express.js, and REST APIs, implementing authentication, product management, and admin functionalities.
- Integrated Cash on Delivery (COD) and prepaid order workflows using Shiprocket APIs, enabling seamless order fulfillment and shipment tracking.
- Built and integrated Razorpay payment gateway using a modular API architecture, implementing secure payment verification, webhook handling, and transaction management.

Projects

Gym Reservation Bot | Python, Selenium, Google Cloud Console

January 2021

- Developed an automatic bot using Python and Google Cloud Console to register myself for a timeslot at my school gym.
- Implemented Selenium to create an instance of Chrome in order to interact with the correct elements of the web page.
- Created a Linux virtual machine to run on Google Cloud so that the program is able to run everyday from the cloud.
- Used Cron to schedule the program to execute automatically at 11 AM every morning so a reservation is made for me.

PrepPilot | Java, Android Studio

March 2026

- Developed PrepPilot, an interview preparation platform featuring DSA sheets, aptitude practice, project idea generator.
- Built progress tracking and personalized learning workflows to help users organize and monitor interview preparation.
- Integrated AI-powered assistance and role-based preparation roadmaps to provide guided learning and support.
- Added an open-source contribution hub with workflow explanations, beginner guides, and curated repositories.

DocQuery AI | Python, FastAPI, ChromaDB, BM25, Streamlit

January 2026

- Built a RAG-based platform for document search, summarization, and comparison across multiple document formats.
- Integrated Gemini AI with OCR-powered document ingestion, citation tracking, and real-time response streaming.
- Designed a Hybrid RAG pipeline using ChromaDB, BM25, and RRF ranking to deliver accurate context retrieval and semantic search.

Technical Skills

Languages: Java, JavaScript, Python, SQL

Frameworks: React.js, Next.js, Node.js, Express.js, Flask, REST APIs, Tailwind CSS

AI Technologies: RAG, LangChain, Gemini API

Tools: Git, GitHub, AWS, Jira Postman, Render

Databases: MongoDB, MySQL

Leadership / Extracurricular

GirlScript Summer of Code (GSSoC)

May 2026 – Present

Project Admin

Remote

- Managed PrepPilot AI as a Project Admin, mentoring contributors and guiding open-source development activities.
- Reviewed pull requests, triaged issues, and coordinated feature development to maintain project quality.
- Facilitated contributor onboarding through documentation, contribution guides, and community support.